Top genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
TRIM15	−6.982156	1.744110e−11	2.795e−07	4.481e−03
SPEGNB	6.654981	1.700017e−10	1.362e−06	1.092e−02
DNAH14	−6.497850	4.888554e−10	2.612e−06	1.395e−02
FGD5	6.284854	1.968977e−09	7.888e−06	2.228e−02
BIRC3	−6.250128	2.460702e−09	7.888e−06	2.228e−02
CNTF	6.098956	6.405819e−09	1.251e−05	2.228e−02
DCAF12L1	6.084222	7.023496e−09	1.251e−05	2.228e−02
BTBD18	6.108801	6.022947e−09	1.251e−05	2.228e−02
FAR2	6.148465	4.694191e−09	1.251e−05	2.228e−02
TTC9C	5.920001	1.931640e−08	2.756e−05	3.563e−02
ZFAND4	5.929166	1.826864e−08	2.756e−05	3.563e−02
LDLRAD4	5.909148	2.063290e−08	2.756e−05	3.563e−02
FBXO42	5.845898	3.023061e−08	3.488e−05	3.563e−02
CTNND1	5.844611	3.046517e−08	3.488e−05	3.563e−02
GPR179	−5.831599	3.293929e−08	3.520e−05	3.563e−02
HIRA	5.809944	3.749632e−08	3.756e−05	3.563e−02
GYS2	5.795250	4.093171e−08	3.859e−05	3.563e−02
NUP160	5.779558	4.493833e−08	4.002e−05	3.563e−02
ZNF311	−5.645636	9.874276e−08	8.330e−05	6.951e−02
ATP10A	5.629839	1.082268e−07	8.673e−05	6.951e−02
ATP6V0A2	5.601534	1.274776e−07	9.730e−05	7.426e−02
RBSN	5.504094	2.226429e−07	1.552e−04	1.081e−01
SLC13A5	−5.508978	2.165540e−07	1.552e−04	1.081e−01
FBXO24	5.406465	3.856848e−07	2.473e−04	1.585e−01
MARCHF8	5.406839	3.848804e−07	2.473e−04	1.585e−01

Top genes by Q−Value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
TRIM15	−6.982156	1.744110e−11	2.795e−07	4.481e−03
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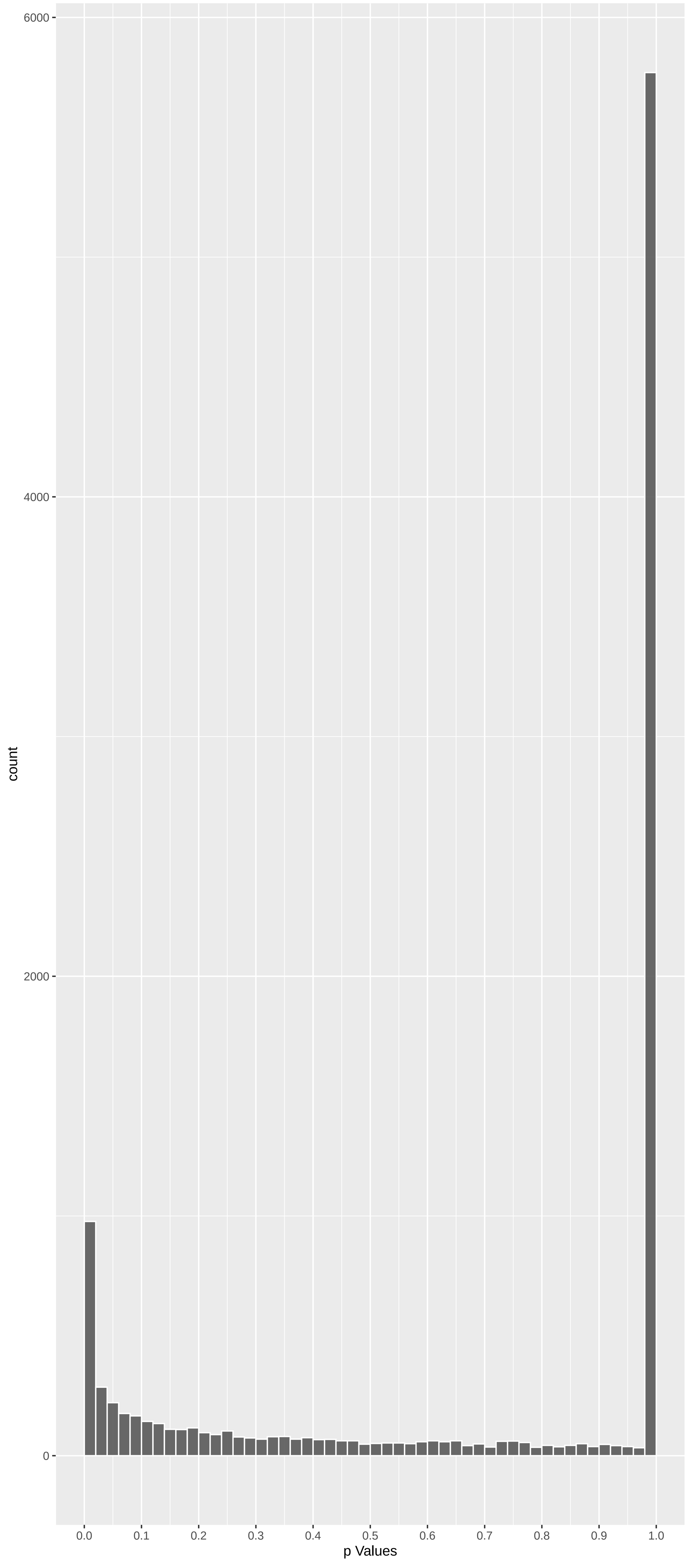
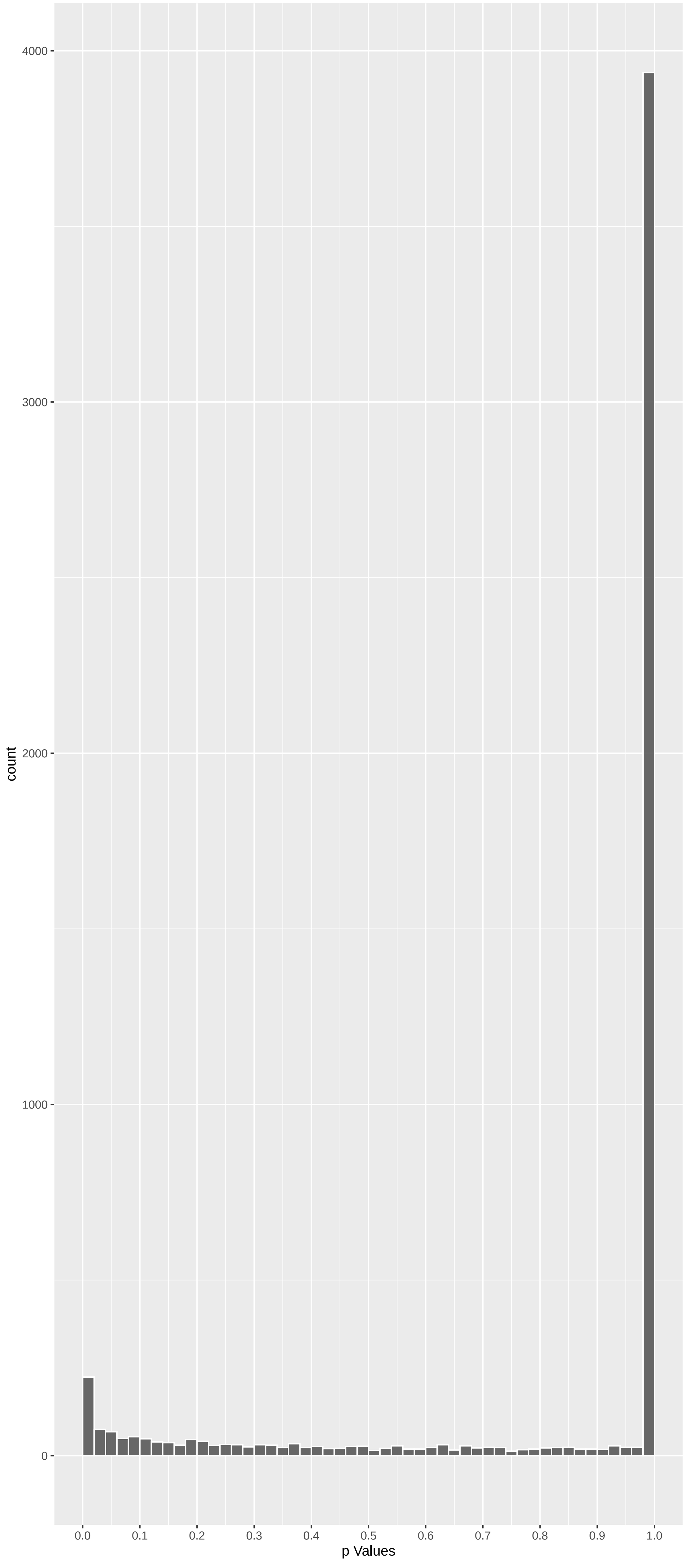
DNAH14	−6.497850	4.888554e−10	2.612e−06	1.395e−02
CNTF	6.098956	6.405819e−09	1.251e−05	2.228e−02
DCAF12L1	6.084222	7.023496e−09	1.251e−05	2.228e−02
BTBD18	6.108801	6.022947e−09	1.251e−05	2.228e−02
FAR2	6.148465	4.694191e−09	1.251e−05	2.228e−02
FGD5	6.284854	1.968977e−09	7.888e−06	2.228e−02
BIRC3	−6.250128	2.460702e−09	7.888e−06	2.228e−02
GYS2	5.795250	4.093171e−08	3.859e−05	3.563e−02
HIRA	5.809944	3.749632e−08	3.756e−05	3.563e−02
TTC9C	5.920001	1.931640e−08	2.756e−05	3.563e−02
ZFAND4	5.929166	1.826864e−08	2.756e−05	3.563e−02
LDLRAD4	5.909148	2.063290e−08	2.756e−05	3.563e−02
FBXO42	5.845898	3.023061e−08	3.488e−05	3.563e−02
NUP160	5.779558	4.493833e−08	4.002e−05	3.563e−02
CTNND1	5.844611	3.046517e−08	3.488e−05	3.563e−02
GPR179	−5.831599	3.293929e−08	3.520e−05	3.563e−02
ATP10A	5.629839	1.082268e−07	8.673e−05	6.951e−02
ZNF311	−5.645636	9.874276e−08	8.330e−05	6.951e−02
ATP6V0A2	5.601534	1.274776e−07	9.730e−05	7.426e−02
RBSN	5.504094	2.226429e−07	1.552e−04	1.081e−01
SLC13A5	−5.508978	2.165540e−07	1.552e−04	1.081e−01
FBXO24	5.406465	3.856848e−07	2.473e−04	1.585e−01
MARCHF8	5.406839	3.848804e−07	2.473e−04	1.585e−01

Positive Rho Non−permulated



Negative Rho Non−permulated



Top Positive genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
SPEGNB	6.654981	1.700017e−10	1.362e−06	1.092e−02
FGD5	6.284854	1.968977e−09	7.888e−06	2.228e−02
CNTF	6.098956	6.405819e−09	1.251e−05	2.228e−02
DCAF12L1	6.084222	7.023496e−09	1.251e−05	2.228e−02
BTBD18	6.108801	6.022947e−09	1.251e−05	2.228e−02
FAR2	6.148465	4.694191e−09	1.251e−05	2.228e−02
TTC9C	5.920001	1.931640e−08	2.756e−05	3.563e−02
ZFAND4	5.929166	1.826864e−08	2.756e−05	3.563e−02
LDLRAD4	5.909148	2.063290e−08	2.756e−05	3.563e−02
FBXO42	5.845898	3.023061e−08	3.488e−05	3.563e−02

Top Negative genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
TRIM15	−6.982156	1.744110e−11	2.795e−07	4.481e−03
DNAH14	−6.497850	4.888554e−10	2.612e−06	1.395e−02
BIRC3	−6.250128	2.460702e−09	7.888e−06	2.228e−02
GPR179	−5.831599	3.293929e−08	3.520e−05	3.563e−02
ZNF311	−5.645636	9.874276e−08	8.330e−05	6.951e−02
SLC13A5	−5.508978	2.165540e−07	1.552e−04	1.081e−01
PIP	−5.270141	8.179145e−07	4.370e−04	2.049e−01
PLPP1	−5.245923	9.330135e−07	4.465e−04	2.049e−01
CFHR5	−5.195932	1.222182e−06	5.237e−04	2.209e−01
N4BP2L2	−5.183226	1.308480e−06	5.378e−04	2.210e−01

GO_Biological_Process_2023 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Inorganic Cation Transmembrane Transport	0.09203326	277	1.548e−07	8.339e−04	ATP6V0A2:16 KCNK18:30 ATP2A1:47 KCNA4:114 SLC6A6:115 KCNQ5:125
Monoatomic Cation Transmembrane Transpor	0.08776057	273	6.773e−07	1.825e−03	ATP6V0A2:16 KCNK18:30 ATP2A1:47 KCNA4:114 SLC6A6:115 KCNQ5:125
Carboxylic Acid Transport (GO:0046942)	0.16850755	50	3.803e−05	5.122e−02	SLC43A1:68 SLC6A6:115 SLC6A15:192 SLC36A2:358 SLC6A19:402 SLC6A14:588
Potassium Ion Transmembrane Transport (G	0.10567398	132	2.872e−05	5.122e−02	KCNK18:30 KCNJ16:53 KCNA4:114 KCNQ5:125 LRRC55:159 KCNJ1:213
Monocarboxylic Acid Transport (GO:001571	0.14653168	63	5.854e−05	6.158e−02	SLC27A4:37 TSPO2:126 ATP8B1:176 SLC10A2:250 MFSD2A:341 ABCD3:374
Nitrogen Compound Transport (GO:0071705)	0.09409977	151	6.857e−05	6.158e−02	SLC43A1:68 SLC6A6:115 SLC22A4:153 RD3:174 SLC6A15:192 SLC22A1:240
NADH Dehydrogenase Complex Assembly (GO:	−0.16308881	48	9.366e−05	6.308e−02	NDUFA11:192 NDUFB10:305 NDUFA8:471 TMEM126B:500 NDUFB8:505 OXA1L:630
Mitochondrial Respiratory Chain Complex	−0.16308881	48	9.366e−05	6.308e−02	NDUFA11:192 NDUFB10:305 NDUFA8:471 TMEM126B:500 NDUFB8:505 OXA1L:630
Axoneme Assembly (GO:0035082)	−0.20440312	30	1.073e−04	6.398e−02	CCDC40:29 TTC12:302 RSPH6A:417 RP1:477 CFAP74:695 SPAG1:842
Fatty Acid Transport (GO:0015908)	0.18292934	37	1.187e−04	6.398e−02	SLC27A4:37 CD36:187 SLC22A1:240 FABP9:317 MFSD2A:341 FABP1:472
Oxidative Phosphorylation (GO:0006119)	−0.14448858	55	2.125e−04	1.041e−01	ATP5PD:113 NDUFA11:192 NDUFB10:305 NDUFV3:360 ATP5PB:450 ATP5MG:452
Defense Response To Bacterium (GO:004274	−0.10003435	113	2.456e−04	1.103e−01	NLRP1:32 DEFB132:150 IL18:355 RNASE8:490 TLR6:498 AZU1:588
Fatty Acid Catabolic Process (GO:0009062	0.13686037	58	3.157e−04	1.212e−01	SLC27A4:37 ECI2:79 LPIN2:137 DECR1:189 ABCD3:374 ACOT8:524
One−Carbon Compound Transport (GO:001975	0.19238848	29	3.375e−04	1.212e−01	SLC4A3:112 SLC4A1:146 SLC4A7:323 CA2:392 SLC14A2:408 UPK3A:544
Proton Motive Force−Driven Mitochondrial	−0.15151399	47	3.290e−04	1.212e−01	ATP5PD:113 NDUFA11:192 NDUFB10:305 NDUFV3:360 ATP5PB:450 ATP5MG:452
Cilium−Dependent Cell Motility (GO:00602	−0.24787225	17	4.035e−04	1.359e−01	DNAH2:36 TEKT1:320 CFAP54:548 DNAH3:554 DNAH1:677 DNAH8:739
Amino Acid Transport (GO:0006865)	0.15246805	43	5.456e−04	1.633e−01	SLC43A1:68 SLC6A6:115 SLC6A15:192 SLC36A2:358 SLC6A19:402 SLC6A14:588
piRNA Processing (GO:0034587)	0.23572407	18	5.366e−04	1.633e−01	GPAT2:49 ANKRD34B:104 TDRKH:389 PNLDC1:514 FKBP6:687 DDX4:954
Gamma−Aminobutyric Acid Transport (GO:00	0.39932000	6	7.056e−04	2.001e−01	SLC6A6:115 SLC6A1:768 SLC6A12:964 SLC6A11:1806 SLC6A13:2838 SLC9A3R1:3008
Amino Acid Import Across Plasma Membrane	0.19320087	25	8.292e−04	2.185e−01	SLC1A1:64 SLC6A6:115 TSPO2:126 SLC22A4:153 SLC6A14:588 SLC38A5:653
Peptide Biosynthetic Process (GO:0043043	−0.08726391	122	8.923e−04	2.185e−01	MRPL24:154 GGT6:255 MRPL37:388 CARNS1:400 DMD:445 PABPC4:517
Potassium Ion Transport (GO:0006813)	0.08837742	119	8.886e−04	2.185e−01	KCNK18:30 KCNJ16:53 KCNA4:114 KCNQ5:125 LRRC55:159 KCNH4:214
Fatty Acid Beta−Oxidation (GO:0006635)	0.13785216	47	1.085e−03	2.436e−01	ECI2:79 GCDH:127 DECR1:189 ABCD3:374 ACOT8:524 PEX2:797
Regulation Of Type II Interferon Product	−0.11111668	73	1.042e−03	2.436e−01	PTPN22:167 IL18:355 RIPK3:432 CD14:448 CD244:513 TLR3:713
Cilium Movement (GO:0003341)	−0.13684630	47	1.180e−03	2.543e−01	CCDC40:29 DNAI2:80 CFAP251:160 TEKT1:320 RSPH6A:417 CFAP54:548
Mitochondrial Respiratory Chain Complex	−0.10646916	77	1.254e−03	2.599e−01	NDUFA11:192 NDUFB10:305 LYRM7:405 NDUFA8:471 TMEM126B:500 NDUFB8:505
T Cell Mediated Immunity (GO:0002456)	−0.24490531	14	1.512e−03	2.733e−01	C4BPA:488 EMP2:612 BTN3A3:734 CD8A:913 CD70:997 MYO1G:1021
Aerobic Respiration (GO:0009060)	−0.12408686	54	1.623e−03	2.733e−01	NDUFA11:192 NDUFB10:305 NDUFV3:360 NDUFA8:471 NDUFB8:505 OXA1L:630
Defense Response To Symbiont (GO:0140546	−0.09090314	101	1.623e−03	2.733e−01	NLRP1:32 CD207:35 NLRP9:64 CD40:237 MX2:278 RIPK3:432
Long−Chain Fatty Acid Transport (GO:0015	0.19127927	23	1.500e−03	2.733e−01	SLC27A4:37 CD36:187 FABP9:317 MFSD2A:341 ABCD3:374 PPARG:889
Proton Motive Force−Driven ATP Synthesis	−0.12921419	51	1.423e−03	2.733e−01	ATP5PD:113 NDUFA11:192 NDUFB10:305 NDUFV3:360 ATP5PB:450 ATP5MG:452
Very Long−Chain Fatty Acid Catabolic Pro	0.37293333	6	1.559e−03	2.733e−01	SLC27A4:37 ABCD3:374 ACOX1:1527 ABCD2:2568 SLC27A2:3576 ABCD4:3898
Metal Ion Transport (GO:0030001)	0.07246803	157	1.775e−03	2.876e−01	KCNK18:30 CLDN16:45 ATP2A1:47 KCNJ16:53 KCNA4:114 TRPM7:169
Positive Regulation Of Cytokine Producti	−0.05519815	272	1.815e−03	2.876e−01	TRIM15:1 CCL1:38 BRCA1:51 NLRP9:64 NFAM1:71 IL6ST:117
Protein Stabilization (GO:0050821)	−0.06677090	182	1.952e−03	3.004e−01	PPIB:188 CPN2:212 PHB2:268 APOA2:294 STK3:357 MDM4:429
Sodium Ion Transmembrane Transport (GO:0	0.10125709	78	2.015e−03	3.015e−01	SLC6A6:115 SLC6A15:192 SCNN1G:263 NALCN:377 SLC6A19:402 SLC12A3:531
Regulation Of Cilium Beat Frequency (GO:	−0.39556835	5	2.189e−03	3.188e−01	CCDC40:29 CYB5D1:398 CFAP43:847 CFAP206:853 DNAH11:6452 NA
Centromere Complex Assembly (GO:0034508)	−0.24406899	13	2.314e−03	3.196e−01	MIS12:383 DLGAP5:595 HJURP:685 CENPC:738 MIS18A:1051 OIP5:1146
Receptor Signaling Pathway Via STAT (GO:	−0.16091162	30	2.294e−03	3.196e−01	CSF2RB:15 GHR:574 STAT2:579 JAK3:663 MYO1C:777 CCR2:821
Neutral Amino Acid Transport (GO:0015804	0.14854804	34	2.733e−03	3.681e−01	SLC1A1:64 SLC43A1:68 SLC6A6:115 SLC6A15:192 SLC36A2:358 SLC6A14:588

EnrichmentHsSymbolsFile2 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_TRANSPORT_OF_INORGANIC_CATIONS_	0.18717674	101	8.300e−11	5.385e−07	SLC1A1:64 SLC43A1:68 SLC4A3:112 SLC6A6:115 SLC4A1:146 SLC6A15:192
REACTOME_SLC_MEDIATED_TRANSMEMBRANE_TRAN	0.09595441	236	4.030e−07	1.307e−03	SLC27A4:37 SLC1A1:64 SLC43A1:68 SLC4A3:112 SLC6A6:115 SLC4A1:146
BENPORATH_ES_WITH_H3K27ME3	0.04455657	978	2.961e−06	6.404e−03	RGS9:20 DUSP8:43 LYSMD2:71 KCNQ5:125 PADI2:150 SNTB1:156
REACTOME_AMINO_ACID_TRANSPORT_ACROSS_THE	0.23208494	31	7.760e−06	1.259e−02	SLC43A1:68 SLC6A6:115 SLC6A15:192 SLC36A2:358 SLC6A19:402 SLC6A14:588
REACTOME_THE_CITRIC_ACID_TCA_CYCLE_AND_R	−0.09962129	149	2.762e−05	2.987e−02	ATP5PD:113 LRPPRC:144 LDHC:157 NDUFA11:192 NDUFB10:305 ME2:313
REACTOME_SLC_TRANSPORTER_DISORDERS	0.12893282	90	2.391e−05	2.987e−02	NUP160:14 SLC27A4:37 SLC1A1:64 SLC4A1:146 SLC40A1:222 NUP210:238
NIKOLSKY_BREAST_CANCER_5P15_AMPLICON	0.25083793	22	4.648e−05	3.764e−02	SLC6A19:402 LPCAT1:504 CEP72:904 AHRR:975 SDHA:1104 EXOC3:1164
FLORIO_HUMAN_NEOCORTEX	−0.33750187	12	5.158e−05	3.764e−02	FCRL4:87 FUT6:108 ACSM2B:524 FAM177B:577 PIWIL3:769 ERAP2:987
REACTOME_CHROMOSOME_MAINTENANCE	−0.11894923	97	5.221e−05	3.764e−02	CENPU:82 ATRX:174 POLR2H:215 MIS18BP1:472 HJURP:685 TERF1:688
REACTOME_DEFENSINS	−0.26378101	19	6.878e−05	4.119e−02	DEFB132:150 TLR1:166 DEFB136:204 CCR2:821 DEFB129:1005 CCR6:1049
REACTOME_TRANSPORT_OF_SMALL_MOLECULES	0.04572783	658	6.983e−05	4.119e−02	ATP10A:15 ATP6V0A2:16 ALB:25 SLC27A4:37 ATP2A1:47 ABCA5:54
MEISSNER_NPC_HCP_WITH_H3K4ME2_AND_H3K27M	0.06233550	330	1.040e−04	5.625e−02	ST6GAL2:66 ANKRD34B:104 FOXF2:172 GHSR:193 FFAR4:201 TBX21:248
REACTOME_BETA_DEFENSINS	−0.27498560	16	1.400e−04	6.486e−02	DEFB132:150 TLR1:166 DEFB136:204 CCR2:821 DEFB129:1005 CCR6:1049
REACTOME_NA_CL_DEPENDENT_NEUROTRANSMITTE	0.25353156	19	1.304e−04	6.486e−02	SLC6A6:115 SLC6A15:192 SLC22A1:240 SLC6A19:402 SLC6A14:588 SLC6A1:768
MIKKELSEN_MCV6_HCP_WITH_H3K27ME3	0.05271389	419	2.266e−04	8.871e−02	FAR2:3 PODXL2:23 SKOR1:69 ANK1:144 MYO16:181 TBX21:248
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT_	−0.10680445	101	2.105e−04	8.871e−02	ATP5PD:113 LRPPRC:144 NDUFA11:192 NDUFB10:305 NDUFV3:360 ATP5PB:450
WEBER_METHYLATED_HCP_IN_SPERM_DN	0.20856871	26	2.324e−04	8.871e−02	GTF2A1L:452 TULP2:457 STK31:458 SOX30:526 DMRT1:688 RBMXL2:710
REACTOME_DEPOSITION_OF_NEW_CENPA_CONTAIN	−0.16945241	39	2.511e−04	9.051e−02	CENPU:82 MIS18BP1:472 HJURP:685 CENPC:738 CENPS:746 H2AC14:1012
REACTOME_SPERM_MOTILITY_AND_TAXES	−0.37177847	8	2.709e−04	9.251e−02	CATSPERB:33 CATSPER3:197 CATSPER4:491 CATSPERG:963 CATSPERD:976 CATSPER2:1040
BENPORATH_PRC2_TARGETS	0.04443151	577	2.854e−04	9.257e−02	LYSMD2:71 GHSR:193 FFAR4:201 RIMBP3B:229 DCC:230 TBX21:248
REACTOME_ANTIMICROBIAL_PEPTIDES	−0.16260762	41	3.158e−04	9.314e−02	DEFB132:150 TLR1:166 DEFB136:204 RNASE8:490 CCR2:821 ELANE:922
WP_MITOCHONDRIAL_COMPLEX_I_ASSEMBLY_MODE	−0.15213421	47	3.092e−04	9.314e−02	NDUFB10:305 NDUFV3:360 NDUFA8:471 TMEM126B:500 NDUFB8:505 TMEM70:628
MIKKELSEN_MEF_HCP_WITH_H3K27ME3	0.04243617	553	6.859e−04	1.810e−01	SKOR1:69 CRACR2B:80 PADI2:150 SPRN:161 DLEU7:185 RUNDC3A:191
REACTOME_ERYTHROCYTES_TAKE_UP_OXYGEN_AND	0.39721210	6	7.525e−04	1.810e−01	SLC4A1:146 CA2:392 CA1:1071 RHAG:1725 CA4:1815 AQP1:4732
REACTOME_CYTOKINE_SIGNALING_IN_IMMUNE_SY	−0.04032492	607	7.407e−04	1.810e−01	BIRC3:3 CSF2RB:15 TNFRSF6B:21 GH2:45 NEDD4:53 LCK:63
REACTOME_REPRODUCTION	−0.10132104	95	6.485e−04	1.810e−01	CATSPERB:33 BRCA1:51 BRCA2:137 CATSPER3:197 CATSPER4:491 MLH3:502
WOO_LIVER_CANCER_RECURRENCE_DN	0.11908825	67	7.534e−04	1.810e−01	SARDH:31 GCDH:127 MAT1A:140 MTTP:371 BPHL:527 ACADSB:621
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT	−0.10801874	81	7.827e−04	1.814e−01	LRPPRC:144 NDUFA11:192 NDUFB10:305 NDUFV3:360 NDUFA8:471 TMEM126B:500
RODRIGUES_THYROID_CARCINOMA_POORLY_DIFFE	0.03740577	694	8.472e−04	1.895e−01	CTNND1:11 HIRA:12 NT5C3B:57 ABI2:59 SLC1A1:64 ST6GAL2:66
SERVITJA_LIVER_HNF1A_TARGETS_UP	0.08696931	121	9.633e−04	1.925e−01	MGST3:353 PCDH20:462 PDK4:602 BUB1:603 CYP7A1:609 PSAT1:647
REACTOME_COMPLEX_I_BIOGENESIS	−0.13756652	48	9.792e−04	1.925e−01	NDUFA11:192 NDUFB10:305 NDUFV3:360 NDUFA8:471 TMEM126B:500 NDUFB8:505
WP_PROXIMAL_TUBULE_TRANSPORT	0.13691501	49	9.174e−04	1.925e−01	SLC1A1:64 SLC36A2:358 CA2:392 SLC6A19:402 SLC5A8:415 SLC34A1:974
YOSHIMURA_MAPK8_TARGETS_UP	0.02976382	1104	9.578e−04	1.925e−01	GYS2:13 RGS9:20 CLPS:22 PNLIPRP1:41 KCNJ16:53 SLC1A1:64
REACTOME_METABOLISM_OF_LIPIDS	0.03723986	677	1.027e−03	1.960e−01	FAR2:3 ALB:25 MORC2:33 GPAT2:49 ECI2:79 MOGAT2:89
ACEVEDO_METHYLATED_IN_LIVER_CANCER_DN	0.03830330	634	1.066e−03	1.976e−01	FGD5:2 SLC43A1:68 OTUD4:91 PPP1R12B:97 NCOA4:132 GRAMD1C:133
REACTOME_INHIBITION_OF_DNA_RECOMBINATION	−0.16281104	32	1.437e−03	2.590e−01	ATRX:174 POLR2H:215 TERF1:688 TINF2:887 POLR2G:992 H2AC14:1012
REACTOME_FERTILIZATION	−0.19737035	21	1.742e−03	3.055e−01	CATSPERB:33 CATSPER3:197 CATSPER4:491 ADAM21:564 CATSPERG:963 CATSPERD:976
REACTOME_MICRORNA_MIRNA_BIOGENESIS	−0.20067040	20	1.892e−03	3.230e−01	POLR2H:215 AGO4:644 POLR2G:992 BCDIN3D:1132 POLR2F:1291 XPO5:1315
NIKOLSKY_BREAST_CANCER_7P22_AMPLICON	0.14524016	38	1.951e−03	3.245e−01	BRAT1:78 SUN1:499 TMEM184A:516 FBXL18:534 PAPOLB:1283 UNCX:1410
GRANDVAUX_IRF3_TARGETS_DN	0.22123736	16	2.185e−03	3.394e−01	TIMP3:1032 TYRP1:1301 SORL1:1457 ANXA4:1878 GEM:1920 RASA1:2013

DisGeNET Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Hypercalciuria	0.17696804	44	4.923e−05	2.896e−01	CLDN16:45 TRPV6:186 KCNJ1:213 PTH:218 WNK4:354 PRPF4B:381
Nephrocalcinosis	0.15812849	54	5.904e−05	2.896e−01	CLDN16:45 SDS:134 SLC4A1:146 KCNJ1:213 PTH:218 HOGA1:284
Pediatric Obesity	0.13251960	65	2.229e−04	7.290e−01	VEGFA:334 AMY1C:364 LEP:417 ACAD8:469 MCHR2:729 GHRL:739
Anemia, hereditary spherocytic hemolytic	0.43537186	5	7.474e−04	7.497e−01	SPTA1:38 ANK1:144 SLC4A1:146 SPTB:242 EPB42:4601 NA
22q11 Deletion Syndrome	0.22869581	19	5.595e−04	7.497e−01	HIRA:12 ARVCF:277 DGCR8:503 SNAI1:1078 SEC24C:1678 COMT:1974
22q11 partial monosomy syndrome	0.30012234	11	5.678e−04	7.497e−01	HIRA:12 ARVCF:277 PI4KA:894 SEC24C:1678 COMT:1974 RREB1:2357
Asymmetric crying face association	0.26446634	13	9.621e−04	7.497e−01	HIRA:12 CASP3:205 ARVCF:277 SEC24C:1678 COMT:1974 RREB1:2357
Atherogenesis	0.14015211	44	1.305e−03	7.497e−01	ESR1:265 NPC1:288 VEGFA:334 ALOX5:380 PTGS2:443 HRH1:616
Bardet−Biedl Syndrome	0.13545290	55	5.162e−04	7.497e−01	VEGFA:334 LEP:417 LZTFL1:612 IQCB1:643 AZIN1:824 TMEM67:1144
Distal Renal Tubular Acidosis	0.24468953	14	1.526e−03	7.497e−01	ALB:25 SLC4A1:146 CA2:392 OSGEP:746 SLC26A7:1170 EPHA3:1681
Electroretinogram abnormal	0.08085884	130	1.487e−03	7.497e−01	RNF216:27 RD3:174 PROM1:224 CRB1:347 KCNJ13:368 KIZ:444
Female infertility	0.17207436	28	1.629e−03	7.497e−01	STAG3:81 ESR1:265 LEP:417 ESR2:1022 LEPR:1108 NR1I2:1111
Hyperactive renin−angiotensin system	0.33228547	8	1.136e−03	7.497e−01	KCNJ1:213 SCNN1G:263 SLC12A1:810 SCNN1B:911 CLCNKB:2079 NR3C2:3065
Increased plasma renin activity	0.33228547	8	1.136e−03	7.497e−01	KCNJ1:213 SCNN1G:263 SLC12A1:810 SCNN1B:911 CLCNKB:2079 NR3C2:3065
Kidney Calculi	0.13241761	53	8.610e−04	7.497e−01	CLDN16:45 SLC4A1:146 TRPM7:169 PTH:218 LRPAP1:359 CLCN5:468
Mitochondrial Diseases	−0.04922224	346	1.758e−03	7.497e−01	LRPPRC:144 ASPM:187 NDUFA11:192 ATAD3B:195 C3:253 NDUFB10:305
Olivopontocerebellar Atrophies	0.21151936	19	1.416e−03	7.497e−01	SLC1A1:64 CASP3:205 ERCC6:484 GRIA2:634 GRIA3:738 GRM3:818
Polydipsia	0.17368184	28	1.473e−03	7.497e−01	CLDN16:45 KCNJ1:213 LEP:417 SLC12A3:531 CDC73:595 CASR:796
Retinal Diseases	0.05182931	313	1.706e−03	7.497e−01	CNTF:5 RGS9:20 ALB:25 SLC6A6:115 RD3:174 DECR1:189
Schizophrenia	0.02504194	1614	1.123e−03	7.497e−01	CNTF:5 TTC9C:8 RGS9:20 SLC1A1:64 MROH2A:107 TMX2:109
Seborrheic dermatitis	0.19825982	21	1.663e−03	7.497e−01	HIRA:12 ARVCF:277 NFE2L2:1354 SLCO2A1:1566 SEC24C:1678 BTD:1856
Spherocytosis	0.43537186	5	7.474e−04	7.497e−01	SPTA1:38 ANK1:144 SLC4A1:146 SPTB:242 EPB42:4601 NA
Tetany	0.22381624	21	3.854e−04	7.497e−01	HIRA:12 CLDN16:45 KCNJ1:213 PTH:218 ARVCF:277 SLC12A3:531
Sterility, Postpartum	0.17599337	24	2.847e−03	7.852e−01	ESR1:265 LEP:417 ESR2:1022 LEPR:1108 NR1I2:1111 ADIPOR2:1307
Subfertility, Female	0.17599337	24	2.847e−03	7.852e−01	ESR1:265 LEP:417 ESR2:1022 LEPR:1108 NR1I2:1111 ADIPOR2:1307
Fanconi−Bickel Syndrome	0.31319912	8	2.158e−03	7.852e−01	ABCC8:862 SLC34A1:974 EHHADH:1651 BTD:1856 AGPAT2:2654 GPX3:2767
Female sterility	0.17599337	24	2.847e−03	7.852e−01	ESR1:265 LEP:417 ESR2:1022 LEPR:1108 NR1I2:1111 ADIPOR2:1307
Hypertensive disease	0.02724750	1104	2.720e−03	7.852e−01	ALB:25 SARDH:31 CLDN16:45 S100A6:95 CYP11A1:123 SDS:134
Idiopathic hypercalciuria	0.33563886	7	2.103e−03	7.852e−01	CLCN5:468 SLC12A3:531 CASR:796 SLC12A1:810 VDR:2371 TRPV5:3780
Metabolic Diseases	0.04657120	361	2.515e−03	7.852e−01	CNTF:5 MOGAT2:89 ENPP2:93 GCDH:127 LPIN2:137 MAT1A:140
Paranasal Sinus Diseases	−0.14205707	39	2.152e−03	7.852e−01	CCDC40:29 PTPN22:167 DRC1:603 ATM:687 FMR1:741 SPAG1:842
Pigmentary iris degeneration	0.17875581	24	2.441e−03	7.852e−01	WHRN:533 PCDH15:632 CDH23:1260 TYRP1:1301 PDZD7:1560 PITX2:1687
Sebaceous Adenocarcinoma	−0.17957165	24	2.331e−03	7.852e−01	BRCA1:51 BRCA2:137 ERBB2:298 TICAM2:704 TMED7:745 WWOX:6452
Transient neonatal diabetes mellitus	0.30426216	8	2.882e−03	7.852e−01	ABCC8:862 INS:938 CDKN1C:1234 EIF2AK3:1608 ZACN:2624 SLC2A2:3360
Unilateral Breast Carcinoma	−0.35518014	6	2.587e−03	7.852e−01	BRCA1:51 BRCA2:137 ERBB2:298 ATM:687 TP53:6452 PALB2:6452
Unilateral Breast Neoplasms	−0.35518014	6	2.587e−03	7.852e−01	BRCA1:51 BRCA2:137 ERBB2:298 ATM:687 TP53:6452 PALB2:6452
Lactic acidemia	−0.07761666	120	3.375e−03	8.488e−01	LRPPRC:144 NDUFA11:192 PPM1B:329 LYRM7:405 TMEM126B:500 SFXN4:549
Metabolic Bone Disorder	0.15508541	30	3.293e−03	8.488e−01	PTH:218 ESR1:265 VEGFA:334 CA2:392 LEP:417 PHEX:490
Occipital myelomeningocele	0.28279312	9	3.306e−03	8.488e−01	HIRA:12 ARVCF:277 SEC24C:1678 COMT:1974 RREB1:2357 UFD1:2683
Abnormality of aortic arch	0.25197915	11	3.808e−03	8.687e−01	HIRA:12 ARVCF:277 SEC24C:1678 COMT:1974 RREB1:2357 UFD1:2683

MGI_Mammalian_Phenotype_Level_4 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MP0001765 abnormal ion homeostasis	0.12259046	103	1.975e−05	9.403e−03	CLDN16:45 CYP11A1:123 SLC4A1:146 TRPV6:186 KCNJ1:213 SCNN1G:263
MP0004043 abnormal pH regulation	0.30902227	14	6.341e−05	1.509e−02	KCNJ1:213 SLC4A7:323 WNK4:354 SLC12A1:810 SCNN1B:911 SLC26A7:1170
MP0000716 abnormal immune system	−0.03979534	968	1.045e−04	1.659e−02	BIRC3:3 CSF2RB:15 SH2D2A:27 CD207:35 MOG:46 BRCA1:51
MP0005636 abnormal mineral homeostasis	0.09539085	134	1.591e−04	1.893e−02	ALB:25 CLDN16:45 CYP11A1:123 ANK1:144 SLC4A1:146 TRPV6:186
MP0002429 abnormal blood cell	−0.03521685	1186	2.120e−04	2.018e−02	BIRC3:3 CSF2RB:15 SH2D2A:27 CD207:35 MOG:46 BRCA1:51
MP0001800 abnormal humoral immune	−0.05416056	326	1.031e−03	5.629e−02	CFD:11 CSF2RB:15 NEDD4:53 LCK:63 C5AR1:81 GON4L:86
MP0003656 abnormal erythrocyte physiolo	0.20002911	22	1.182e−03	5.629e−02	ANK1:144 SLC4A1:146 MAN2A1:345 KCNN4:557 STEAP3:955 NFE2L2:1354
MP0009643 abnormal urine homeostasis	0.06416233	226	1.077e−03	5.629e−02	ALB:25 CLDN16:45 SLC1A1:64 GCDH:127 SLC4A1:146 TRPV6:186
MP0004019 abnormal vitamin homeostasis	0.15069390	41	8.705e−04	5.629e−02	CLDN16:45 TRPV6:186 PHEX:490 CYP7A1:609 RBP2:867 SLC34A1:974
MP0005025 abnormal response to	−0.05329283	346	8.990e−04	5.629e−02	CFD:11 CSF2RB:15 CD207:35 C5AR1:81 IL6ST:117 PML:130
MP0002452 abnormal antigen presenting	−0.03936061	563	2.218e−03	8.122e−02	BIRC3:3 CSF2RB:15 NEDD4:53 LCK:63 GON4L:86 TCIRG1:111
MP0000689 abnormal spleen morphology	−0.04047966	533	2.134e−03	8.122e−02	C5AR1:81 GON4L:86 TCIRG1:111 IL6ST:117 BRCA2:137 PTPN22:167
MP0002723 abnormal immune serum	−0.03800583	622	2.031e−03	8.122e−02	BIRC3:3 CSF2RB:15 SH2D2A:27 MOG:46 NEDD4:53 LCK:63
MP0000230 abnormal systemic arterial	0.07242297	142	3.186e−03	1.052e−01	GYS2:13 WNK4:354 WNK1:428 PTGS2:443 NOX1:485 SLC12A3:531
MP0001819 abnormal immune cell	−0.03161453	857	3.316e−03	1.052e−01	BIRC3:3 CSF2RB:15 SH2D2A:27 MOG:46 BRCA1:51 NEDD4:53
MP0003448 altered tumor morphology	−0.09158068	84	3.923e−03	1.167e−01	BRCA2:137 CXCR3:263 IQGAP1:310 TGM2:541 CCR2:821 CXCR2:850
MP0001756 abnormal urination	0.09121596	80	5.037e−03	1.410e−01	ALB:25 SLC4A1:146 TRPV6:186 KCNJ1:213 SCNN1G:263 SLC14A2:408
MP0002229 neurodegeneration	0.04904178	268	6.727e−03	1.631e−01	CNTF:5 SLC6A6:115 GCDH:127 ANK1:144 RD3:174 ATP8B1:176
MP0002090 abnormal vision	0.12076727	42	6.928e−03	1.631e−01	SLC4A3:112 SLC6A6:115 SLC4A7:323 ELOVL4:464 CYP7A1:609 MERTK:629
MP0002136 abnormal kidney physiology	0.04773929	278	7.270e−03	1.631e−01	ALB:25 CLDN16:45 SLC4A1:146 TRPV6:186 KCNJ1:213 HPX:237
MP0002882 abnormal neuron morphology	0.02899615	875	6.604e−03	1.631e−01	CTNND1:11 ABI2:59 SLC1A1:64 ENPP2:93 SLC4A3:112 SLC6A6:115
MP0002420 abnormal adaptive immunity	−0.02880531	854	7.540e−03	1.631e−01	BIRC3:3 CSF2RB:15 SH2D2A:27 MOG:46 BRCA1:51 NEDD4:53
MP0009642 abnormal blood homeostasis	0.02416074	1246	9.713e−03	1.926e−01	GYS2:13 CLPS:22 ALB:25 SLC27A4:37 CLDN16:45 ATP2A1:47
MP0003953 abnormal hormone level	0.03342164	558	9.667e−03	1.926e−01	GYS2:13 CLDN16:45 FFAR1:58 MOGAT2:89 CYP11A1:123 PNLIP:155
MP0010329 abnormal lipoprotein level	0.08109791	82	1.160e−02	2.209e−01	CLPS:22 MOGAT2:89 CD36:187 ESR1:265 MTTP:371 ALOX5:380
MP0002066 abnormal motor capabilities/c	0.02322213	1074	1.849e−02	3.035e−01	CNTF:5 SLC27A4:37 ABCA5:54 SLC1A1:64 PER2:99 CYP11A1:123
MP0005360 urolithiasis	0.24384745	8	1.698e−02	3.035e−01	SLC26A1:497 SLC2A9:652 AGXT:2220 SLC3A1:2458 SLC7A9:2618 SLC26A6:4089
MP0005365 abnormal bile salt	0.13391575	26	1.830e−02	3.035e−01	ALB:25 PNLIP:155 ATP8B1:176 SLC10A2:250 CYP7A1:609 HSD17B4:1334
MP0000465 gastrointestinal hemorrhage	−0.14053765	24	1.734e−02	3.035e−01	IL6ST:117 IL18:355 FES:449 NFATC1:774 FOXP3:980 ACVRL1:1231
MP0005380 embryogenesis phenotype	−0.03396564	406	2.270e−02	3.274e−01	BRCA1:51 DVL2:75 KLF8:103 GMNN:129 BRCA2:137 ATRX:174
MP0001984 abnormal olfaction	0.14170619	22	2.159e−02	3.274e−01	GJB4:610 BBS1:1385 ID2:1405 RORA:2030 PTPRS:2238 CNGA4:2436
MP0001666 abnormal nutrient absorption	0.10283247	42	2.149e−02	3.274e−01	CLPS:22 ALB:25 MOGAT2:89 PNLIP:155 TRPV6:186 PNLIPRP2:198
MP0001672 abnormal embryogenesis/ devel	−0.03396564	406	2.270e−02	3.274e−01	BRCA1:51 DVL2:75 KLF8:103 GMNN:129 BRCA2:137 ATRX:174
MP0002210 abnormal sex determination	0.03715743	327	2.414e−02	3.380e−01	PIWIL2:32 TAF7L:51 CYP11A1:123 CREM:170 RAD23B:208 ESR1:265
MP0005387 immune system phenotype	−0.06315763	97	3.277e−02	3.609e−01	TCIRG1:111 C3:253 JAK3:663 TICAM2:704 NFATC1:774 AGER:834
MP0005671 abnormal response to	−0.10017540	39	3.085e−02	3.609e−01	CD40:237 CXCR3:263 TNFRSF8:668 BTLA:672 CD8A:913 KIT:1211
MP0001346 abnormal lacrimal gland	0.16439184	14	3.336e−02	3.609e−01	CTNND1:11 PITX3:401 FIGN:1097 E2F1:1364 NCDN:1553 PRLR:2762
MP0001486 abnormal startle reflex	0.06494713	95	2.978e−02	3.609e−01	LATS2:157 CASP3:205 MYO6:223 PCDH15:632 GRIA3:738 SLC6A1:768
MP0000703 abnormal thymus morphology	−0.03564339	338	2.804e−02	3.609e−01	BRCA1:51 LCK:63 TCIRG1:111 IL6ST:117 GMNN:129 BRCA2:137
MP0001919 abnormal reproductive system	0.09757607	40	3.321e−02	3.609e−01	ESR1:265 BAX:383 WT1:391 PTGS2:443 LGR4:569 CXCL12:751

tissue_specific Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
testis	−0.03964168	1380	4.651e−06	2.512e−04	DNAH14:2 SCNN1D:12 ZSCAN5B:19 HEATR4:24 SHBG:25 C1orf100:28
EBV.lymphocyte	−0.04480065	670	1.318e−04	3.558e−03	BIRC3:3 FCRL5:16 ICOSLG:22 BRCA1:51 LCK:63 BRCA2:137
spleen	−0.05358895	298	1.709e−03	3.077e−02	FCRL5:16 FCRL3:20 FCRL6:178 CXCL6:193 FOLR3:203 RENBP:254
cerebellum	0.04259800	374	5.477e−03	7.393e−02	ESPNL:56 SKOR1:69 MTBP:74 PPFIA4:130 MTCL1:141 SPRN:161
bladder	0.15022316	24	1.093e−02	8.999e−02	PADI3:42 RD3:174 UPK3A:544 PSCA:586 UGT1A6:827 PLA2G4A:832
cerebellar.hemisphere	0.03370770	502	1.167e−02	8.999e−02	PODXL2:23 ESPNL:56 MTBP:74 PPFIA4:130 MTCL1:141 ANK1:144
thyroid	0.06074234	150	1.078e−02	8.999e−02	HIRA:12 KCNJ16:53 TMC4:105 FHDC1:279 PGF:296 VEGFA:334
nucleus.accumbens	0.07204380	99	1.368e−02	9.231e−02	ANKRD34B:104 KCNA4:114 HTR6:399 FAT3:440 NWD1:654 PENK:787
breast	0.16217155	15	2.976e−02	1.607e−01	LEP:417 CIDEC:1009 ALX4:1025 HCAR1:2119 LMX1B:2465 NPY2R:2831
Brodmann.area.9	0.04712824	182	2.971e−02	1.607e−01	KCNQ5:125 C3orf80:247 KCNS2:254 NCAM2:416 OLFM1:473 RXFP1:650
putamen	0.15640953	15	3.607e−02	1.771e−01	RGS9:20 KCNH4:214 SLC18A3:678 GRM3:818 OPALIN:2540 ANO3:2742
atrium.auricle	−0.05977775	90	5.095e−02	2.293e−01	FSD2:260 ADAMTS7:290 ASB10:331 HSPB3:514 TNNI3K:531 S100A1:551
substantia.nigra	0.15957272	12	5.573e−02	2.315e−01	KCNE5:1168 FOXB1:1242 SLC6A3:1485 SLC6A11:1806 CHRM5:2186 DBX2:4110
left.ventricle	−0.06311787	68	7.277e−02	2.807e−01	FSD2:260 ASB10:331 HSPB3:514 TNNI3K:531 S100A1:551 ASB18:641
blood	−0.02945221	275	9.716e−02	2.922e−01	CSF2RB:15 SH2D2A:27 NFAM1:71 C5AR1:81 NBEAL2:106 DPEP2:128
C1.spinal.cord	0.03715227	182	8.657e−02	2.922e−01	ENPP2:93 ELOVL7:149 PADI2:150 SEC14L5:270 PLCL1:346 TNFRSF21:385
cortex.kidney	0.04518452	114	9.739e−02	2.922e−01	CLDN16:45 KCNJ1:213 SLCO4C1:306 OGDHL:322 SLC36A2:358 FABP1:472
esophagogastric.junction	−0.17346365	8	8.941e−02	2.922e−01	CHRM3:1296 COL4A5:1609 ADCY5:6452 BARX1:6452 F2RL2:6452 GADL1:6452
ectocervix	0.09088244	25	1.161e−01	3.172e−01	ESR1:265 KRT34:1119 LRFN5:1443 ADRA2A:1454 RIPOR3:1581 ADRA1D:1999
vagina	0.04794107	90	1.175e−01	3.172e−01	SCNN1G:263 ESR1:265 CALML3:455 TMEM40:585 PSCA:586 FAT2:801
pancreas	0.04559552	94	1.282e−01	3.297e−01	CLPS:22 PNLIPRP1:41 SLC43A1:68 PNLIP:155 TRPV6:186 PNLIPRP2:198
transformed.skin.fibroblast	−0.02726735	259	1.357e−01	3.330e−01	NEDD4:53 P4HA3:84 DNAJB4:151 CXCL6:193 SLFN12:264 CHML:274
ovary	0.04106554	102	1.538e−01	3.611e−01	FFAR1:58 SLC4A3:112 TMEM221:400 DLK1:676 BTNL2:734 STAR:861
endocervix	0.04459936	72	1.922e−01	3.843e−01	ESR1:265 PRSS12:694 SP5:759 SNED1:866 IGFBP5:887 ADRA2C:1268
esophagus.mucosa	0.02579425	220	1.920e−01	3.843e−01	PADI3:42 KRT32:92 EPS8L1:235 ADGRF1:275 MAL2:405 CALML3:455
esophagus.muscularis.mucosa	−0.07491134	26	1.866e−01	3.843e−01	PRUNE2:411 KCNMB1:776 COL4A6:1086 CHRM3:1296 CCDC69:1564 F2RL2:6452
prostate	−0.06200804	38	1.867e−01	3.843e−01	CPAMD8:14 CREB3L4:116 TTC6:164 ANO7:766 SHROOM1:814 LMAN1L:1172
subcutaneous.adipose	0.04205296	72	2.188e−01	4.220e−01	CD36:187 FABP9:317 LEP:417 TNMD:489 APOLD1:683 PPARG:889
adrenal.gland	0.02529623	108	3.662e−01	6.818e−01	CYP11A1:123 SNTB1:156 PLEK2:165 SLC40A1:222 SLC14A2:408 MC2R:411
greater.omentum	0.04149454	33	4.101e−01	6.981e−01	TIMP4:147 LEP:417 LPL:893 CIDEC:1009 RBP7:2232 IL6:2605
pituitary.gland	0.01649719	213	4.114e−01	6.981e−01	RD3:174 GHSR:193 CFAP47:200 FFAR4:201 RADX:430 FBXO10:537
skeletal.muscle	−0.01555800	238	4.137e−01	6.981e−01	ALPK3:26 FSD2:260 ART3:292 MSS51:328 ASB10:331 TACC2:412
cerebral.cortex	0.02218227	107	4.302e−01	7.040e−01	CAMK1G:530 DMTN:744 RNF208:997 KCNF1:1114 LRTM2:1125 NPAS1:1314
hypothalamus	0.02555757	66	4.741e−01	7.529e−01	LRRC55:159 PCDH15:632 FOXB1:1242 GRIN3A:1259 GLRA3:1458 B3GAT2:1659
amygdala	0.06461947	9	5.022e−01	7.533e−01	PCDH15:632 NEUROD2:1362 BCAN:2894 LHFPL3:9577 NEUROD6:9577 PTPRZ1:9577
liver	0.01100766	328	5.000e−01	7.533e−01	GYS2:13 ALB:25 SARDH:31 EXOC3L4:44 MOGAT2:89 SDS:134
caudate.nucleus	−0.02497045	26	6.598e−01	8.690e−01	PPP1R1B:1225 AQP4:1255 HS3ST5:6452 CHAT:6452 ETNPPL:6452 FAM237A:6452
hippocampus.proper	−0.04325503	10	6.359e−01	8.690e−01	CABP7:893 NEUROD6:6452 SLC17A7:6452 HRK:6452 OLIG2:6452 GRP:6452
peyers.patch	0.01003718	167	6.570e−01	8.690e−01	MOGAT2:89 SLC10A2:250 CCR9:356 KCNJ13:368 MTTP:371 SLC6A19:402
stomach	0.01478743	86	6.368e−01	8.690e−01	EPN3:163 PSCA:586 GPRC5C:598 CLDN18:718 GHRL:739 SLC9A4:767